Camunda Integration with RASP

After setting up RASP, we need to add the required camunda dependencies to the pom.xml file of RASP.

```
<!-- Camunda BPM Spring Boot Starter -->
<dependency>
    <groupId>org.camunda.bpm.springboot</groupId>
    <artifactId>camunda-bpm-spring-boot-starter-rest</artifactId>
    <version>7.21.0</version> <!-- Replace with the desired version -->
</dependency>

<!-- Camunda BPM Spring Boot Starter Webapp -->
<dependency>
    <groupId>org.camunda.bpm.springboot</groupId>
    <artifactId>camunda-bpm-spring-boot-starter-webapp</artifactId>
    <version>7.21.0</version> <!-- Replace with the desired version -->
</dependency>
```

These are the required camunda dependencies to use camuda 7 diagrams on RASP.

Next change the application.propeties file,

```
server.error.include-message=always
server.port=8081
springfox.documentation.swagger-ui.enabled=true

### Database configuration
#spring.datasource.url=jdbc:h2:file:./camunda-h2-database
## Camunda BPM Admin user configuration
#camunda.bpm.admin-user.id=adi
#camunda.bpm.admin-user.password=adi

## MySQL Configuration
#spring.datasource.url=jdbc:mysql://localhost:3306/camunda
#spring.datasource.username=root
#spring.datasource.password=root
#spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

# Camunda BPM Admin user configuration
camunda.bpm.admin-user.id=admin
camunda.bpm.admin-user.password=password
```

```
# MongoDB Configuration
spring.data.mongodb.uri=mongodb://DB_PASSWORD=root;DB_USERNAME=root;DB_SERVER=localhost:2
3017,/camunda
spring.data.mongodb.database=camunda

## Camunda BPM Admin user configuration
#camunda.bpm.admin-user.id=adi
#camunda.bpm.admin-user.password=adi
```

This creates an admin user in camunda and allows connecting it to MongoDB.

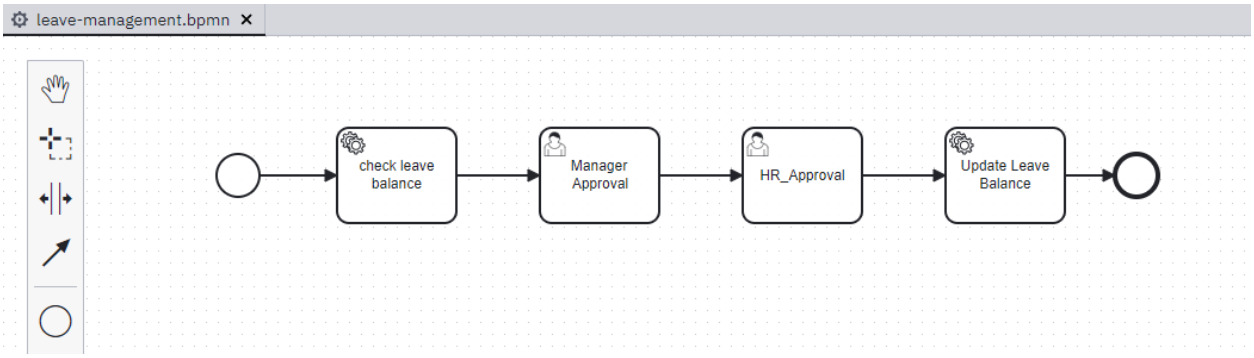
Next, create a Test.java file in the org.rasp.iiitb360 package and paste the following code there.

```
package org.rasp.iiitb360;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class Test {

    public static void main(String... args) {
        SpringApplication.run(Test.class, args);
    }

}
```

Also edit the configuration of the file to add the following in the environment variable,

DB_PASSWORD=root;DB_USERNAME=root;DB_SERVER=localhost:23017,

Next, by using the camunda modeler diagram, create the camunda diagram and save the .bpmn file in the src.main.resources package in the project. Service classes in the bpmn diagram require java classes to work, write the required java classes and place them in the org.rasp.iiitb360 package and paste the path to the file in the java class field of the service task. Remember to set the Type of service class to Java class.



This is a sample Camunda diagram that adds student details entered into mongodb.

```
<?xml version="1.0" encoding="UTF-8"?>
<bpmn:definitions xmlns:bpmn="http://www.omg.org/spec/BPMN/20100524/MODEL"
xmlns:bpmndi="http://www.omg.org/spec/BPMN/20100524/DI"
xmlns:dc="http://www.omg.org/spec/DD/20100524/DC"
xmlns:camunda="http://camunda.org/schema/1.0/bpmn"
xmlns:di="http://www.omg.org/spec/DD/20100524/DI"
xmlns:modeler="http://camunda.org/schema/modeler/1.0" id="Definitions_18imxy7"
targetNamespace="http://bpmn.io/schema/bpmn" exporter="Camunda Modeler"
exporterVersion="5.26.0" modeler:executionPlatform="Camunda Platform"
modeler:executionPlatformVersion="7.21.0">
  <bpmn:process id="Process_1uvp3a3" name="leave" isExecutable="true"
camunda:historyTimeToLive="7">
    <bpmn:extensionElements />
    <bpmn:startEvent id="StartEvent_1">
      <bpmn:extensionElements />
      <bpmn:outgoing>Flow_0s2y304</bpmn:outgoing>
    </bpmn:startEvent>
    <bpmn:serviceTask id="Activity_1jd0owp" name="check leave balance"
camunda:class="org.rasp.iiitb360.LeaveBalanceCheck">
      <bpmn:extensionElements />
      <bpmn:incoming>Flow_0s2y304</bpmn:incoming>
      <bpmn:outgoing>Flow_1h2wzke</bpmn:outgoing>
    </bpmn:serviceTask>
    <bpmn:sequenceFlow id="Flow_0s2y304" sourceRef="StartEvent_1"
targetRef="Activity_1jd0owp" />
    <bpmn:sequenceFlow id="Flow_1h2wzke" sourceRef="Activity_1jd0owp"
targetRef="Activity_0pw3543" />
    <bpmn:userTask id="Activity_0pw3543" name="Manager Approval"
camunda:assignee="manager">
      <bpmn:extensionElements>
        <camunda:formData>
          <camunda:formField id="approve" label="manager approve" type="boolean"
defaultValue="false" />
        </camunda:formData>
      </bpmn:extensionElements>
      <bpmn:incoming>Flow_1h2wzke</bpmn:incoming>
      <bpmn:outgoing>Flow_0y0u36g</bpmn:outgoing>
```

```
    </bpmn:userTask>
    <bpmn:sequenceFlow id="Flow_0y0u36g" sourceRef="Activity_0pw3543"
targetRef="Activity_0h5ifrj" />
    <bpmn:userTask id="Activity_0h5ifrj" name="HR_Approval" camunda:assignee="hr">
      <bpmn:extensionElements>
        <camunda:formData>
          <camunda:formField id="HR_Approval" label="approvehr" type="boolean"
defaultValue="false" />
        </camunda:formData>
      </bpmn:extensionElements>
      <bpmn:incoming>Flow_0y0u36g</bpmn:incoming>
      <bpmn:outgoing>Flow_136a1qj</bpmn:outgoing>
    </bpmn:userTask>
    <bpmn:endEvent id="Event_0gdhtxm">
      <bpmn:incoming>Flow_1gwzdoa</bpmn:incoming>
    </bpmn:endEvent>
    <bpmn:sequenceFlow id="Flow_136a1qj" sourceRef="Activity_0h5ifrj"
targetRef="Activity_0wane8u" />
    <bpmn:sequenceFlow id="Flow_1gwzdoa" sourceRef="Activity_0wane8u"
targetRef="Event_0gdhtxm" />
    <bpmn:serviceTask id="Activity_0wane8u" name="Update Leave Balance"
camunda:class="org.rasp.iiitb360.LeaveBalanceUpdate">
      <bpmn:incoming>Flow_136a1qj</bpmn:incoming>
      <bpmn:outgoing>Flow_1gwzdoa</bpmn:outgoing>
    </bpmn:serviceTask>
  </bpmn:process>
  <bpmndi:BPMNDiagram id="BPMNDiagram_1">
    <bpmndi:BPMNPlane id="BPMNPlane_1" bpmnElement="Process_1uvp3a3">
      <bpmndi:BPMNShape id="_BPMNShape_StartEvent_2" bpmnElement="StartEvent_1">
        <dc:Bounds x="179" y="99" width="36" height="36" />
      </bpmndi:BPMNShape>
      <bpmndi:BPMNShape id="Activity_0x8sz8w_di" bpmnElement="Activity_1jd0owp">
        <dc:Bounds x="280" y="77" width="100" height="80" />
        <bpmndi:BPMNLabel />
      </bpmndi:BPMNShape>
      <bpmndi:BPMNShape id="Activity_1gk5gmo_di" bpmnElement="Activity_0pw3543">
        <dc:Bounds x="450" y="77" width="100" height="80" />
        <bpmndi:BPMNLabel />
      </bpmndi:BPMNShape>
      <bpmndi:BPMNShape id="Activity_0l3mwir_di" bpmnElement="Activity_0h5ifrj">
        <dc:Bounds x="620" y="77" width="100" height="80" />
        <bpmndi:BPMNLabel />
      </bpmndi:BPMNShape>
      <bpmndi:BPMNShape id="Event_0gdhtxm_di" bpmnElement="Event_0gdhtxm">
        <dc:Bounds x="932" y="99" width="36" height="36" />
      </bpmndi:BPMNShape>
      <bpmndi:BPMNShape id="Activity_0b4zvsc_di" bpmnElement="Activity_0wane8u">
        <dc:Bounds x="790" y="77" width="100" height="80" />
        <bpmndi:BPMNLabel />
      </bpmndi:BPMNShape>
```

```
      <bpmndi:BPMNEdge id="Flow_0s2y304_di" bpmnElement="Flow_0s2y304">
        <di:waypoint x="215" y="117" />
        <di:waypoint x="280" y="117" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge id="Flow_1h2wzke_di" bpmnElement="Flow_1h2wzke">
        <di:waypoint x="380" y="117" />
        <di:waypoint x="450" y="117" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge id="Flow_0y0u36g_di" bpmnElement="Flow_0y0u36g">
        <di:waypoint x="550" y="117" />
        <di:waypoint x="620" y="117" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge id="Flow_136a1qj_di" bpmnElement="Flow_136a1qj">
        <di:waypoint x="720" y="117" />
        <di:waypoint x="790" y="117" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge id="Flow_1gwzdoa_di" bpmnElement="Flow_1gwzdoa">
        <di:waypoint x="890" y="117" />
        <di:waypoint x="932" y="117" />
      </bpmndi:BPMNEdge>
    </bpmndi:BPMNPlane>
  </bpmndi:BPMNDiagram>
</bpmn:definitions>
```

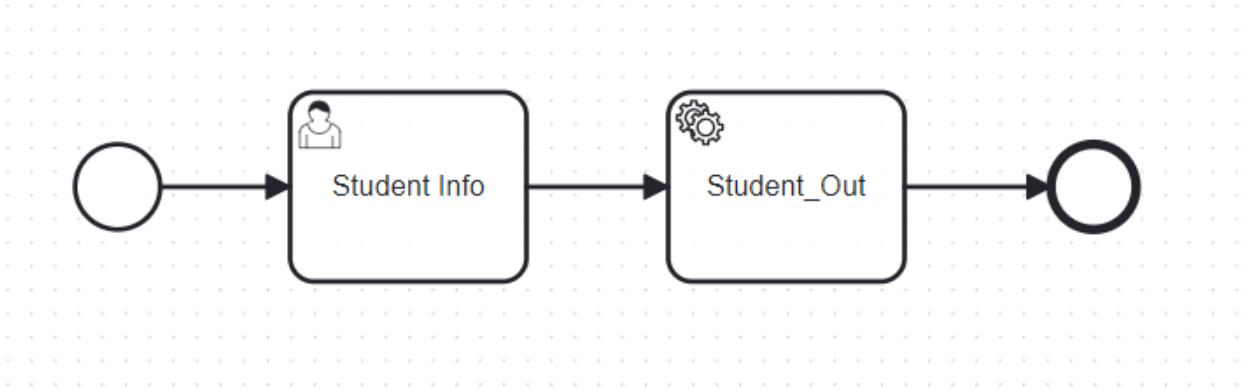
This is what the .bpmn file looks like for the same.

With these steps completed, you can now run camunda in the RASP platform.

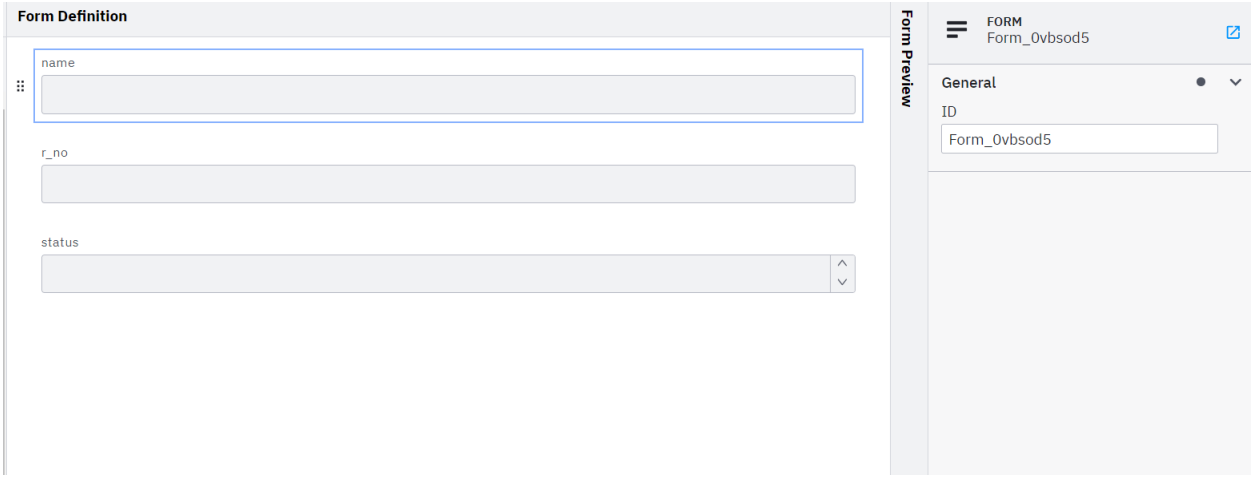
First, run mongodb and connect to port 23017. Then run the Test.java file. Camunda should now be running on localhost:8081 and you are able to add data into the mongodb database.

Now let us take an example of StudentIn – StudentOut where the input of student record can be read from the mongodb and how the data of student status can be stored to mongodb again.

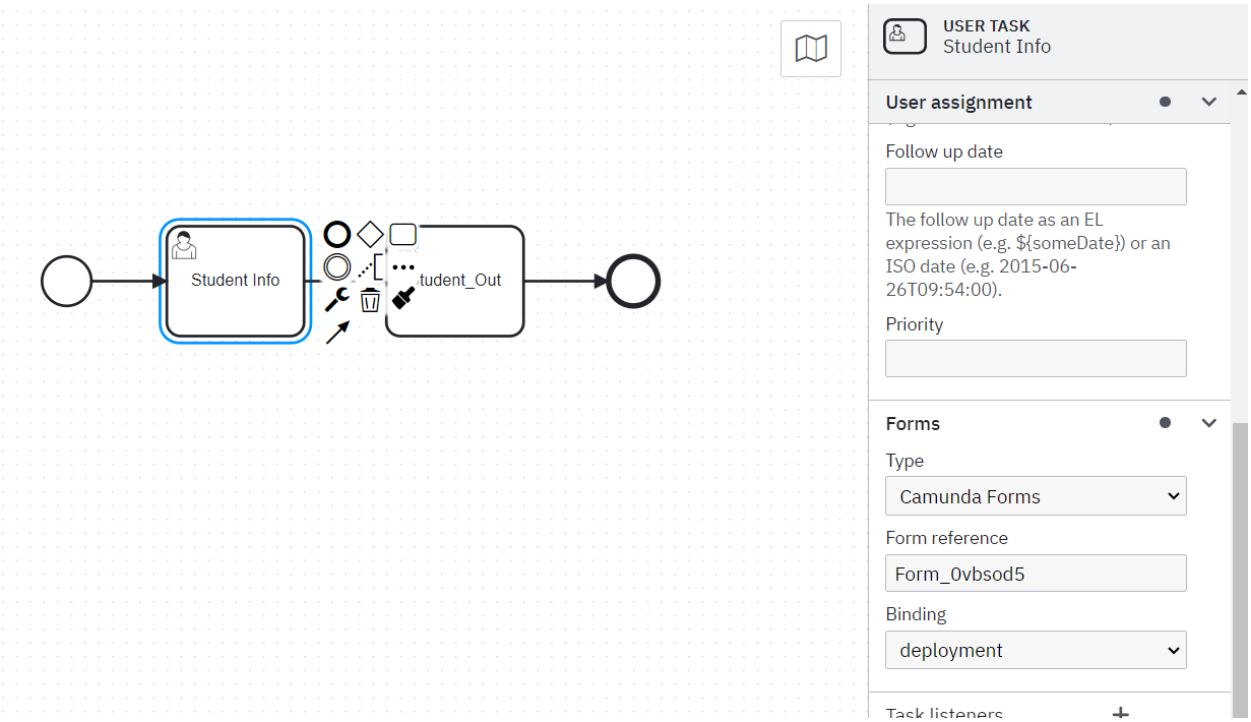
Consider the following Camunda diagram.



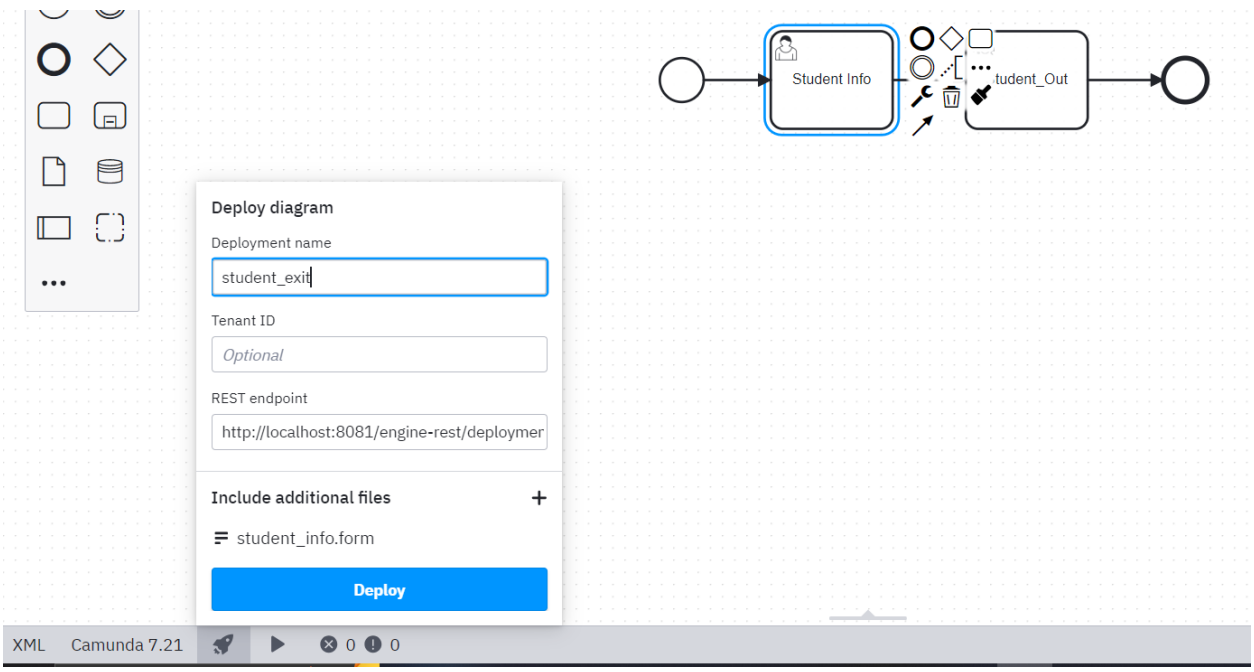
The corresponding student_info.form file for it is



We have to click on the Student_info in .bpmn file and make sure that the Form reference in Forms of the .bpmn file has the ID of our student_info.form

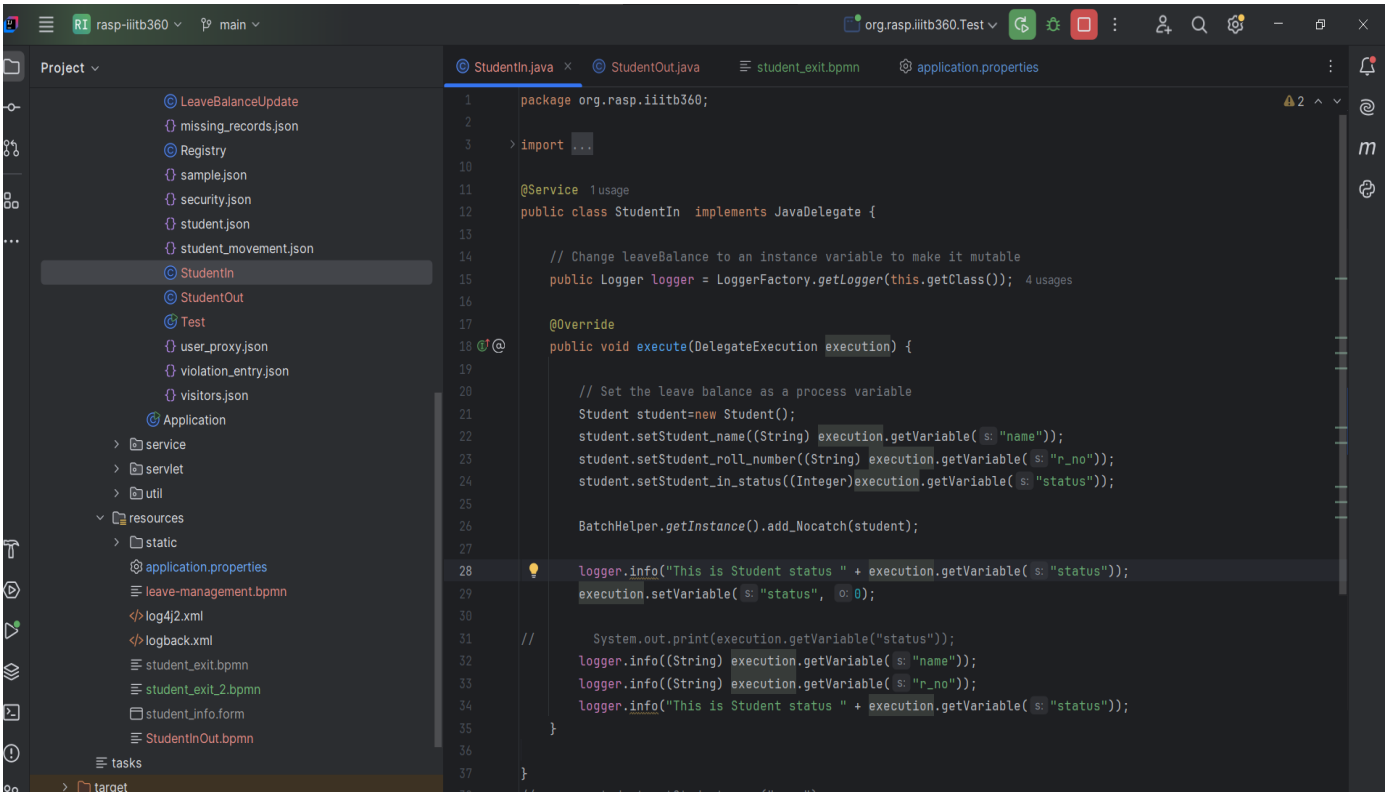


Now let us deploy our camunda file. We have to make sure to add our student_info.form file in “Include additional files “in the Deploy Diagram.



Click on deploy and we get confirmation status. (We have to make sure our Test.java file (a.k.a Camunda_demo.java in the repository) is running when we click deploy).

Setting up StudentIn.java and StudentOut.java



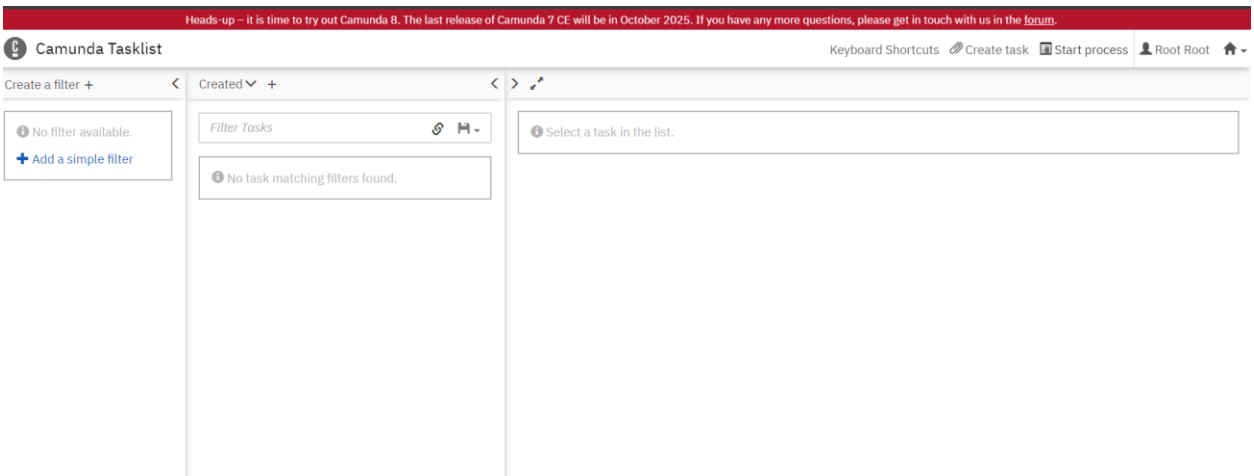
StudentOut.java

```java
@Service  3 usages
public class StudentOut  implements JavaDelegate {

    // Change leaveBalance to an instance variable to make it mutable
    public Logger logger = LoggerFactory.getLogger(this.getClass());  no usages

    @Override
    public void execute(DelegateExecution execution) {

        // Set the leave balance as a process variable
        Student student=new Student();
        String name = (String) execution.getVariable( s: "name");
        String r_no = (String) execution.getVariable( s: "r_no");
        Integer status = (Integer) execution.getVariable( s: "status");


        student.setStudent_name(name);
        student.setStudent_roll_number(r_no);
        student.setStudent_in_status(status);

        execution.setVariable( s: "name", student.getStudent_name());
        execution.setVariable( s: "r_no", student.getStudent_roll_number());
        execution.setVariable( s: "status", student.getStudent_in_status());

        StudentHelper.getInstance().add_Nocatch(student);

        Expression e1 = new Expression( variable: "student_roll_number", REL_OP.EQ,r_no);
        Expression e2 = new Expression(Batch.FIELD_NAME, REL_OP.EQ,name);
        Expression e = Expression.and(e1,e2);
//        BaseResource[] studentRecord = StudentHelper.getInstance().getByExpression(e1);
         BaseResource[] studentRecord = StudentHelper.getInstance().getAll();

        System.out.println(studentRecord.length);
```

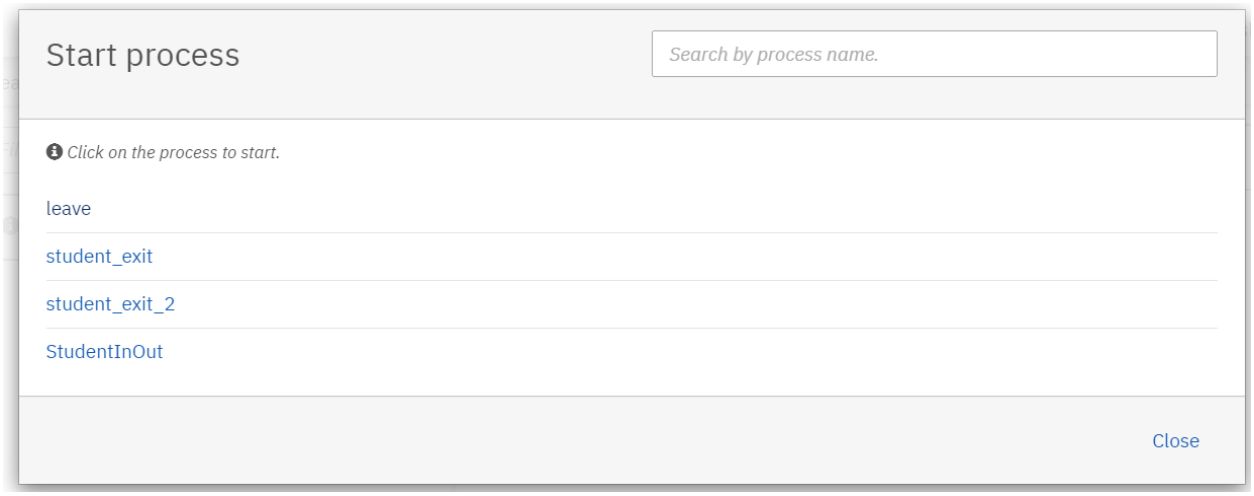
```java
        Expression e1 = new Expression( variable: "student_roll_number", REL_OP.EQ,r_no);
        Expression e2 = new Expression(Batch.FIELD_NAME, REL_OP.EQ,name);
        Expression e = Expression.and(e1,e2);
//        BaseResource[] studentRecord = StudentHelper.getInstance().getByExpression(e1);
         BaseResource[] studentRecord = StudentHelper.getInstance().getAll();

        System.out.println(studentRecord.length);
        System.out.println("Traversing all the student records");
        for (BaseResource baseResource : studentRecord) {
            Student student1 = (Student) baseResource;
            System.out.println("Student Roll Number : " + student1.getStudent_roll_number());
            System.out.println("Student Name : "+ student1.getStudent_name());
        }
```

Note: The add_Nocatch(student) function writes student data into mongodb. Expressions are used to read data from mongodb

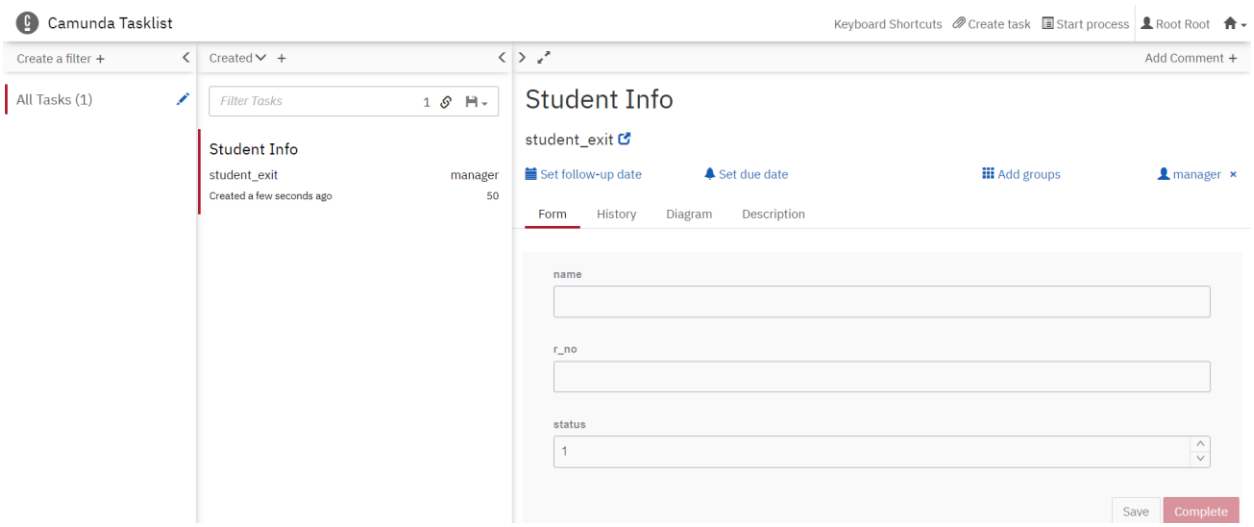
1. Start a process.



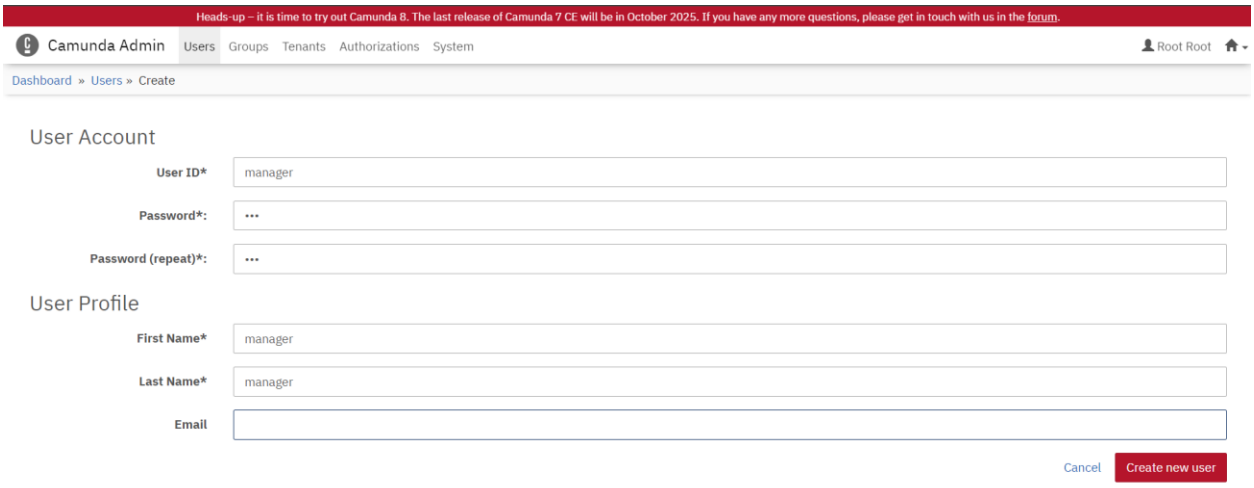
2. Choose the process student_exit



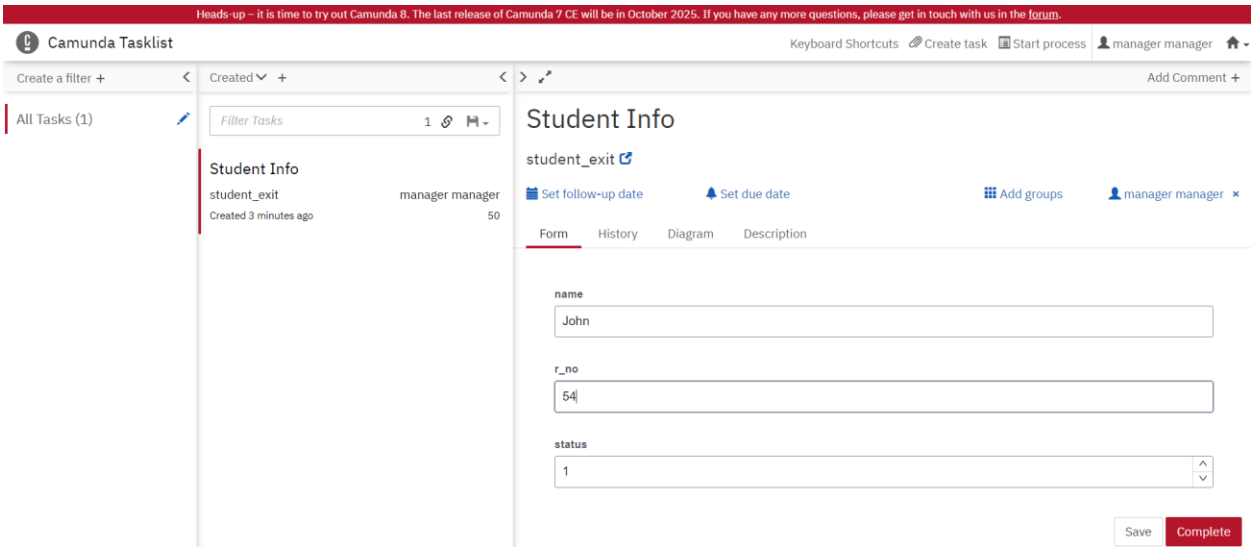
3. Our Camunda tasklist looks as following:



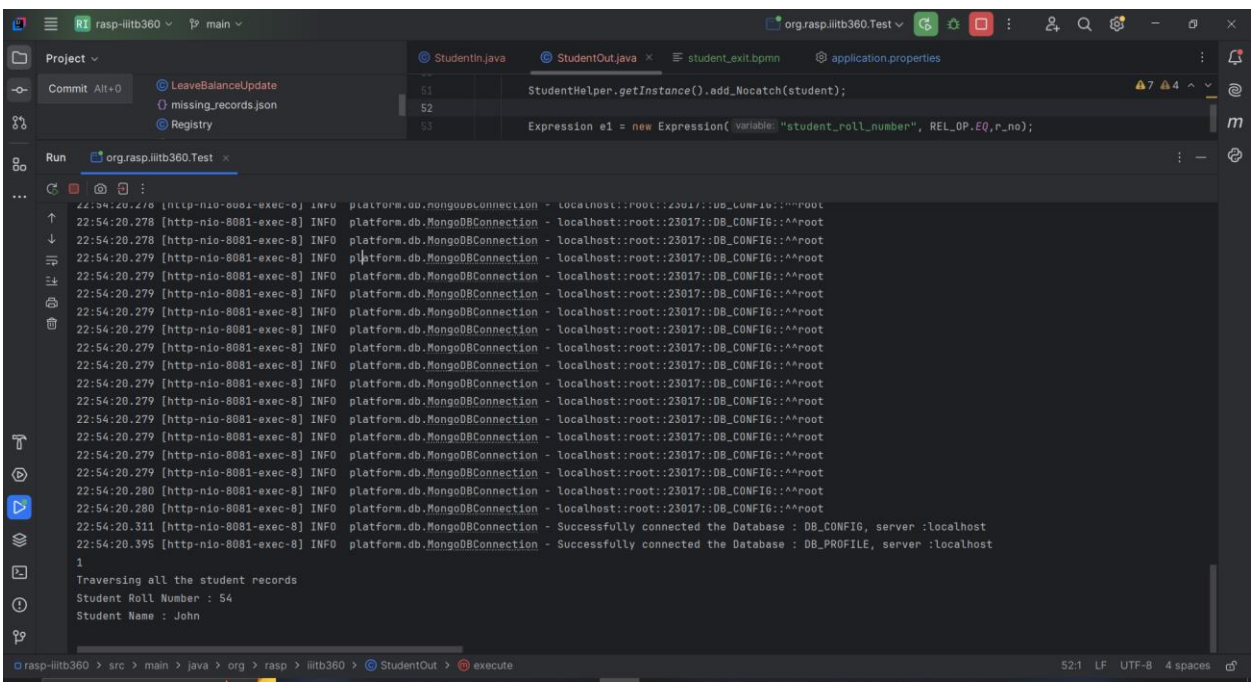
4. Now since the asignee for the state is manager, we will have to create a user called manager and log in as manager. For this create a new user in Camunda with username manager.



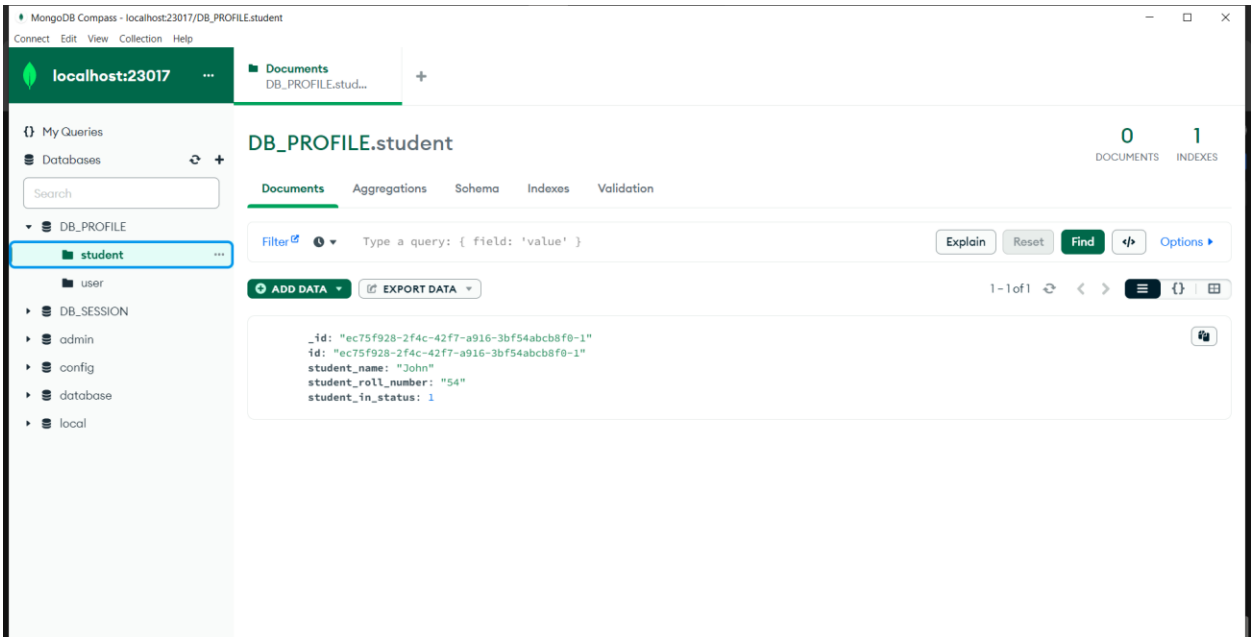
5. Logout from the root account and login as manager and view the tasklist. Fill the form with corresponding student details.



6. We see the following in the terminal



And the following in mongodb


MongoDB Compass - localhost:23017/DB_PROFILE.student
Connect Edit View Collection Help
localhost:23017
Documents
DB_PROFILE.stud...
My Queries
Databases
Search
DB_PROFILE
student
user
DB_SESSION
admin
config
database
local
DB_PROFILE.student
0
DOCUMENTS
1
INDEXES
Documents
Aggregations
Schema
Indexes
Validation
Filter
Type a query: { field: 'value' }
Explain
Reset
Find
Options
ADD DATA
EXPORT DATA
1-1 of 1
_id: "ec75f928-2f4c-42f7-a916-3bf54abcb8f0-1"
id: "ec75f928-2f4c-42f7-a916-3bf54abcb8f0-1"
student_name: "John"
student_roll_number: "54"
student_in_status: 1